AENL226 Power Electronics and power systems (Aug-Dec 2026)

Anandarup Das, Professor

IIT-Delhi Abu Dhabi

anandarup@iitdabudhabi.ac.ae

anandarup@iitd.ac.in

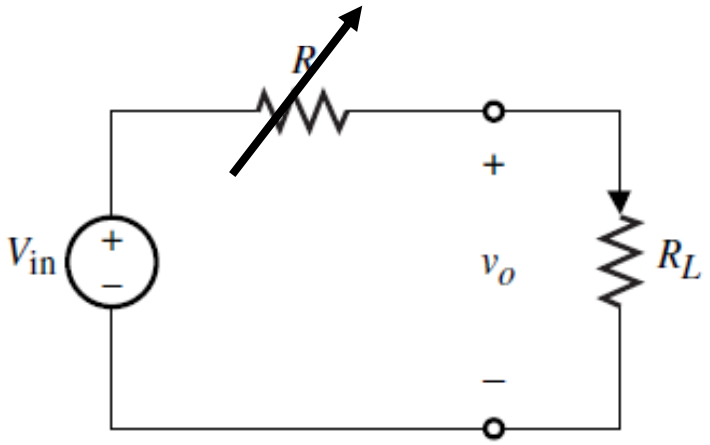
Lecture content

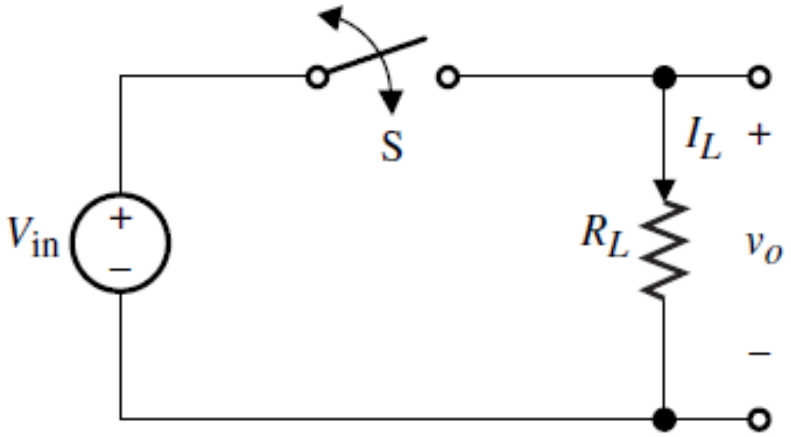
- Why power switches are needed?
- Different types of power switches
- Available switches and their ratings

- See also: Issa Batarseh, chapter 2
- NPTEL lectures

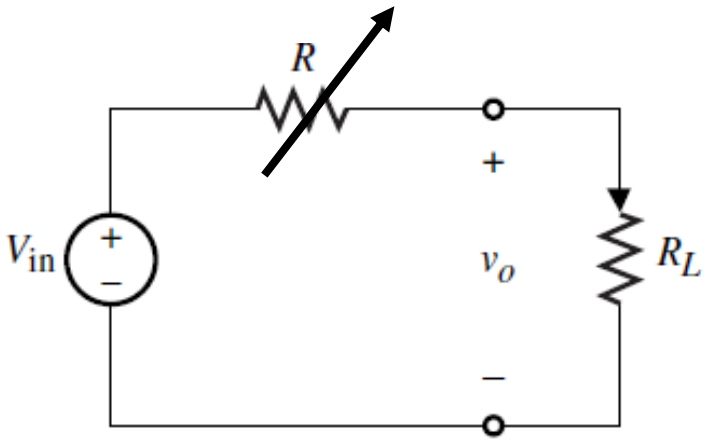
Efficiencies of different types of circuits

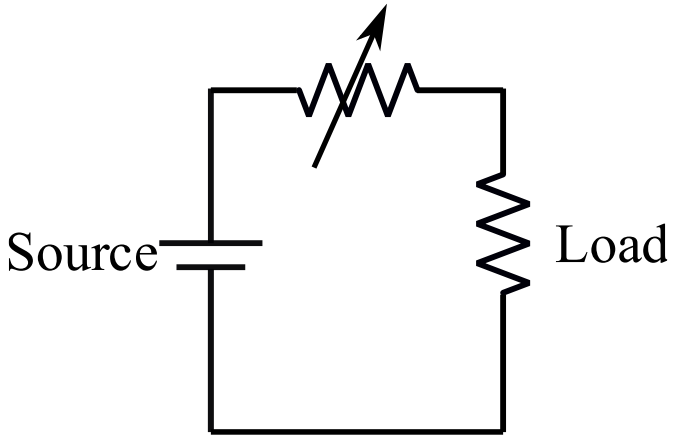
Suppose we have a 24V source and want to deliver 12V output to a 6 ohm resistance. Consider two circuits below. Which has the better efficiency?





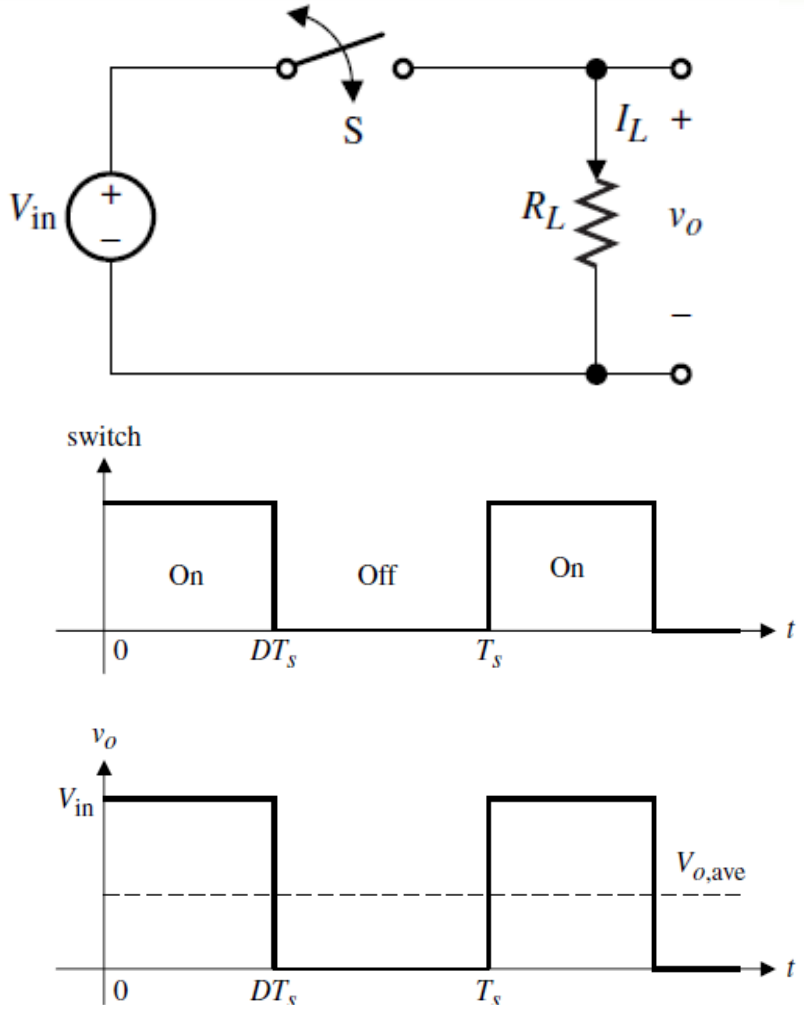
Voltage divider





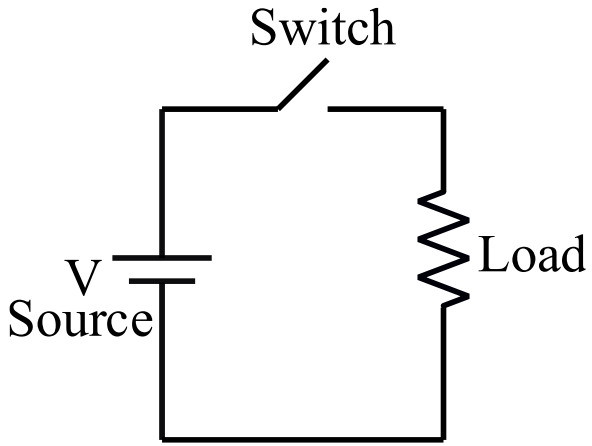
- Assume Vin=24V. Vo=12V. RL=6ohm.
- Current flowing=2A.
- Input power=24V*2A=48W.
- Output power=12*2=24W.
- Efficiency=50%.
- Voltage divider is not a good option.

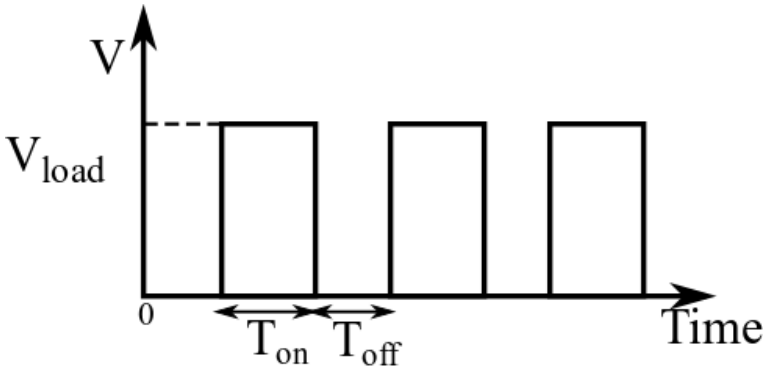
Switching circuit



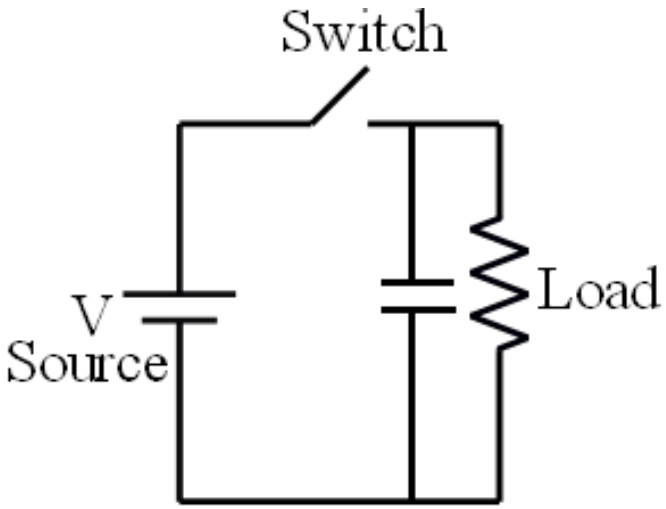
- The switch is turned on and off periodically.
- The output voltage is fluctuating, but its average value is,
  $V_{oavg} = (V_{in} \times T_{on} + 0 \times T_{off})/T_s = D.V_{in}$
- If V_{oavg} = 12V, then D=0.5.
- Output power=input power=24W. Thus we have 100% efficiency (theoretically).
- Challenge: Output voltage is fluctuating, so we need inductors/capacitors to smoothen the transfer of energy.

Introduction of passive components

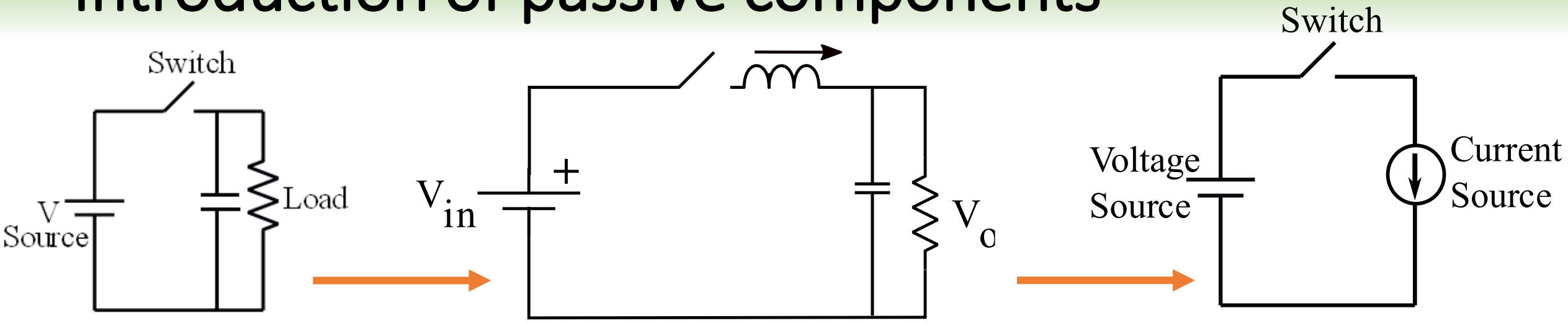




- One drawback of the circuit is the pulsating output voltage.
- We can use a capacitor to minimize the output voltage ripple.

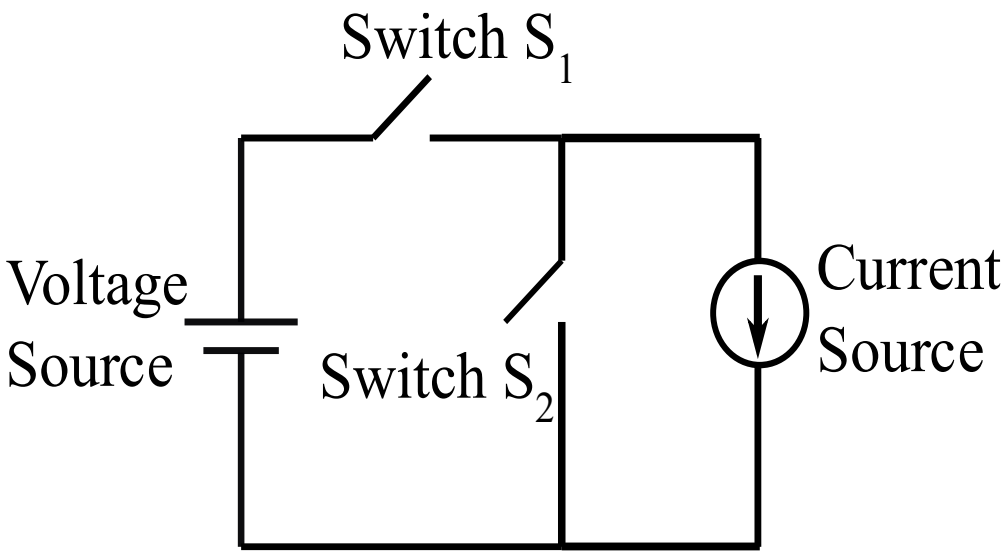


Introduction of passive components



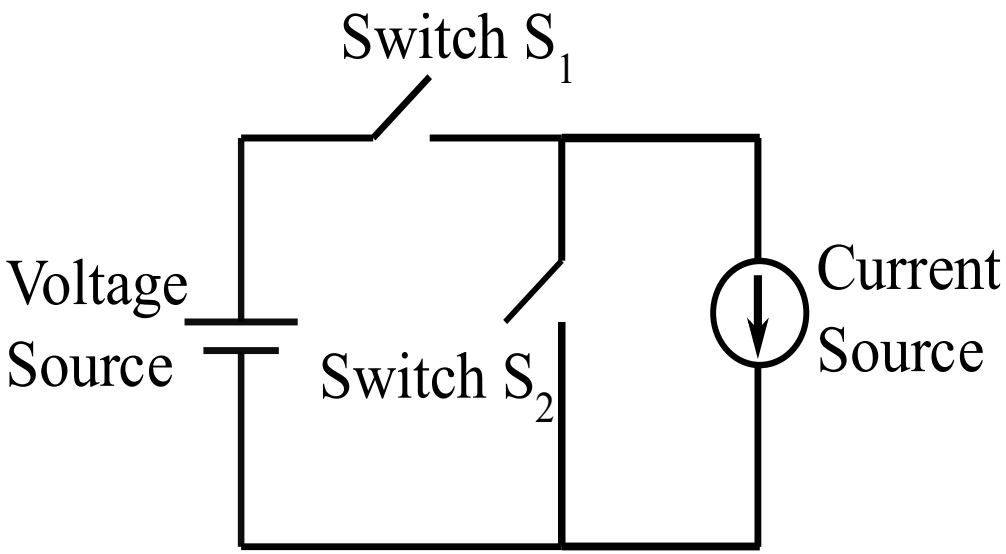
- We also need an inductor; as capacitors will discharge during the off period of the switch. Inductors are used to smoothen the transfer of energy.
- Often loads are inductive in nature e.g., motors, transformers etc.
- Inductive loads are represented as current source.

Current source



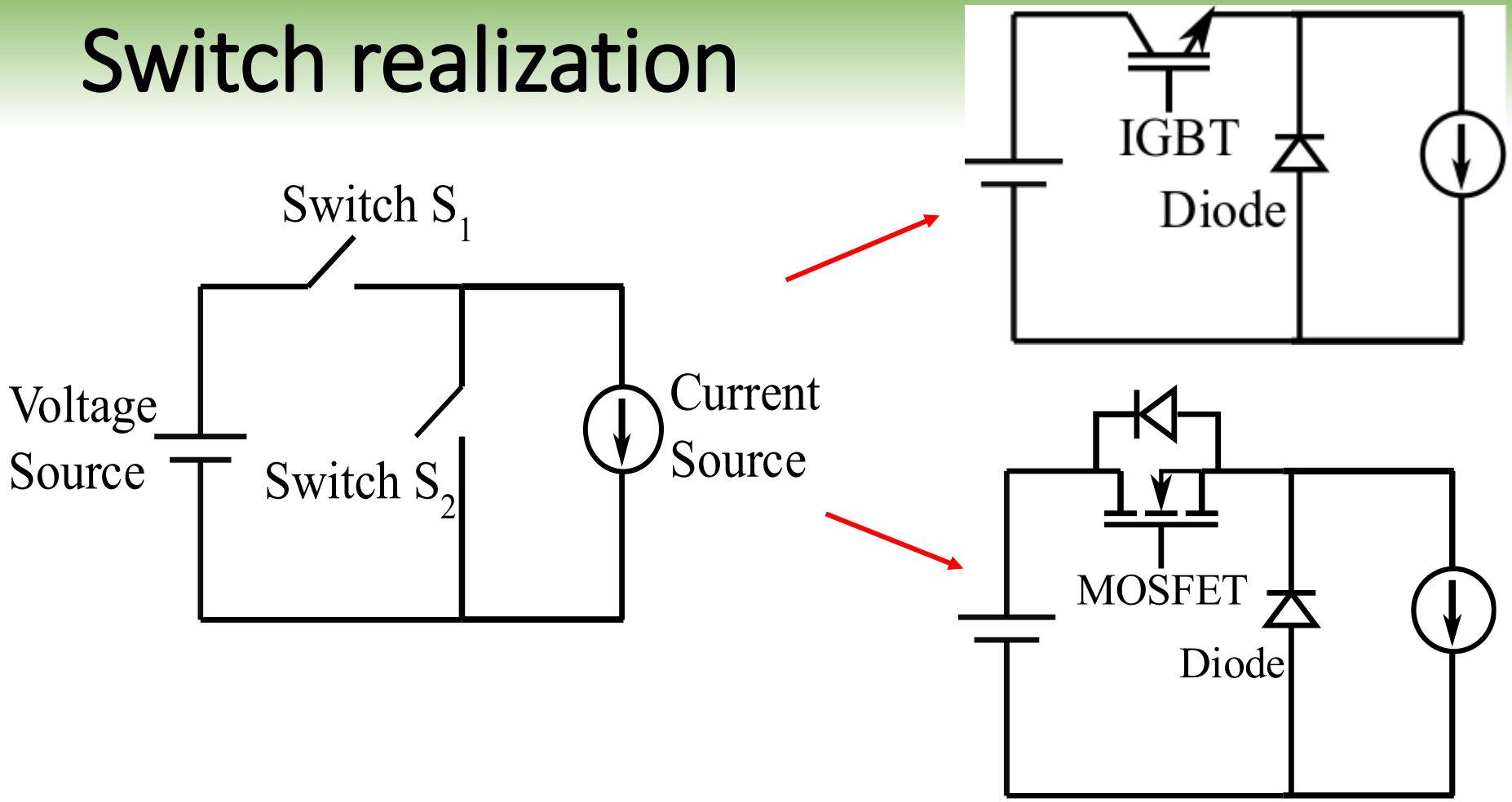
- Current source and voltage source are dual in nature.
- Current sources can be shorted but they cannot be opened.
- Voltage sources can be opened but they cannot be shorted.
- Therefore, we need two switches,
  - Switch S_1 opens the voltages source.
  - Switch S_2 shorts the current source.

Current source



- These two switches can't be operated simultaneously, i.e., if at any time one switch will be ON, then other will be OFF.
- If they are ON at the same time, the voltage source will be shorted.

Switch realization



- There can be many ways to realize the two switches.
- IGBT or MOSFET together with a diode can be used to realize the two switches.

Why to use power switches?

- Switches are used to control power/energy flow from source to load or vice versa.
- Power switches increase efficiency, resistances are avoided.
- Extra losses means bigger heat sinks or extra cooling arrangement. Temperature rise causes lifetime reduction of components.
- Power switches are used as either in ON or OFF states, unlike linear amplifiers.
- Power switches are always switched at a fast rate. They can be switched millions of times in their lifetime. In these aspects, they are better than mechanical switches.
- Disadvantage: Increased complex systems.

ANANDARUP DAS, IITD-AD. FOR INTERNAL USE ONLY, NOT TO BE COPIED.

Properties of ideal power switches

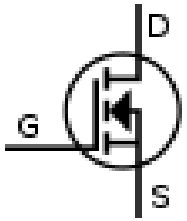
- An ideal switch
  - Can block infinite voltage and conducts zero current when OFF
  - Has zero voltage across and can conduct infinite current when ON
  - Has zero turn on and zero turn off times
  - Has zero power losses and can tolerate infinite temperature in the device

Practical switch

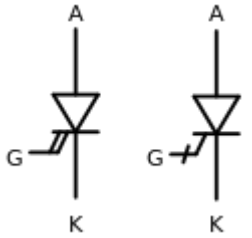
- A practical switch
  - Can block finite voltage (e.g., 100V, 1200V, 1700V etc.)
  - Has some on state voltage drop (e.g., 1-2V) and finite current carrying capacity (e.g., 50A, 1000A, 3000A etc.)
  - Has finite turn on and finite turn off times (e.g., several tens of ns to few us)
  - Maximum junction temperature is limited (e.g., 150/175 degree C)
  - Requires a gate driver (except for diodes)

Practical power switches

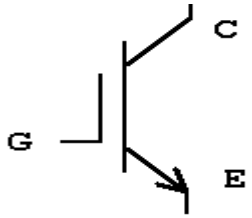
- MOSFET
- IGBT
- IGCT
- GTO
- Thyristor- semi controlled
- Diode-uncontrolled



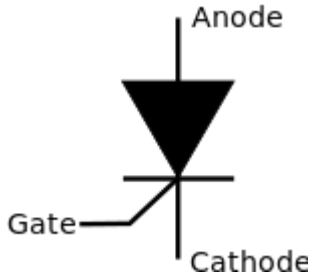
MOSFET



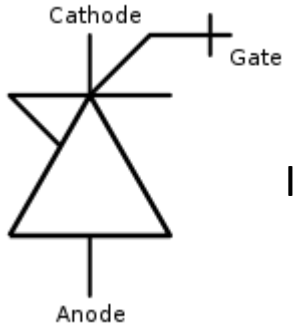
GTO



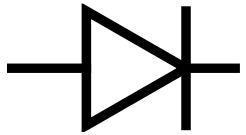
IGBT



Thyristor



IGCT



Diode

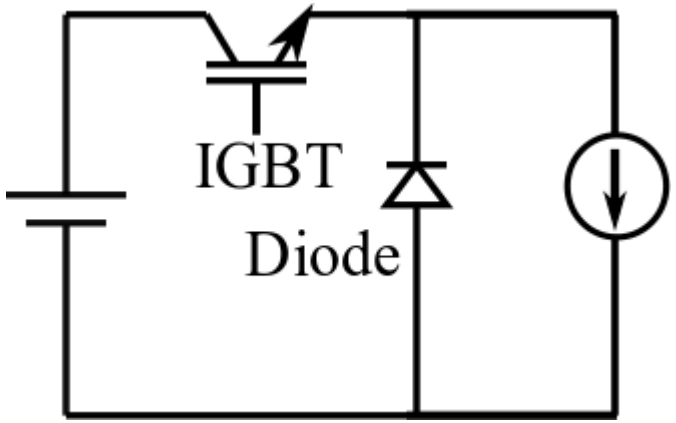
Power semiconductor development

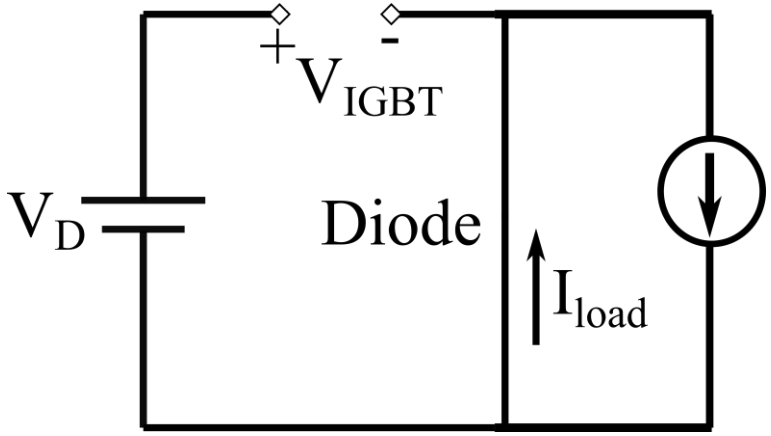
Devices	Voltage Rating	Current Rating	Switching Frequency	Application
Thyristor	~ 7-10 kV	3.5 kA	50-60 Hz	HVDC, getting obsolete
GTO	~6 kV	6kA	1kHZ	HVDC, High power drives, not popular
IGBT (Si)	~6.5kV	3kA	100 kHZ	MV and LV drives, MV converters, Multilevel converter
MOSFET(Si)	600V	~300-400 A	200 kHz	LV drives, LV power supplies, Electric vehicles, Consumer Electronics
MOSFET(SiC)	~1.2kV	~660-700 A	1 MHz	LV drives, Electric vehicles
GaN	~ 900V	~ 50A	10 MHz	Commercial deployment started

*The table changes frequently.

ANANDARUP DAS, IITD-AD. FOR INTERNAL USE ONLY, NOT TO BE COPIED.

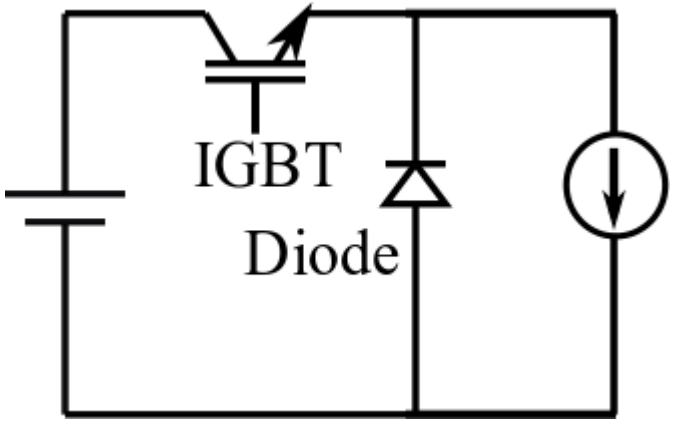
Switch rating

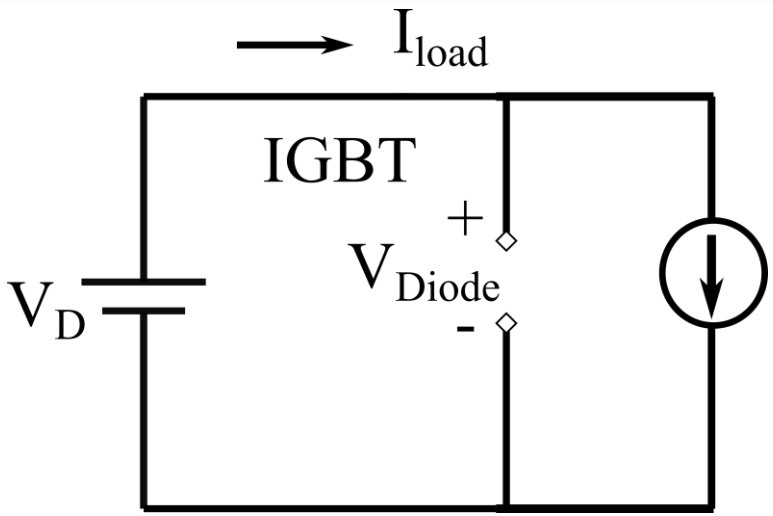




- Let us take the case of IGBT realization.
- The voltage rating of the switch is defined as the voltage across the switch when it is OFF. The current rating of the switch is the current flowing through the switch when it is ON.
- In this case, voltage rating of IGBT is equal to source voltage, $V_{IGBT} = V_D$. The current rating is equal to load current.

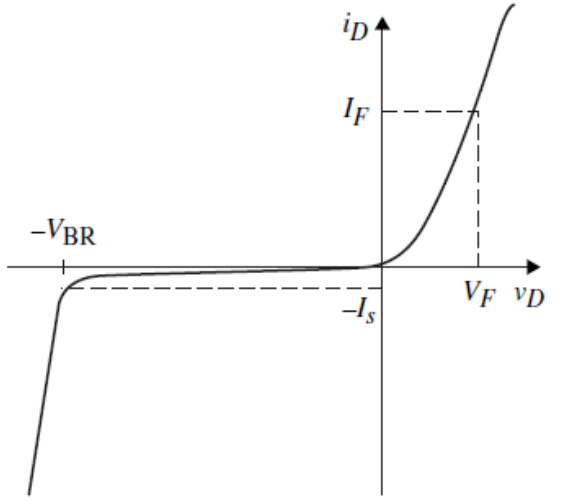
Switch rating

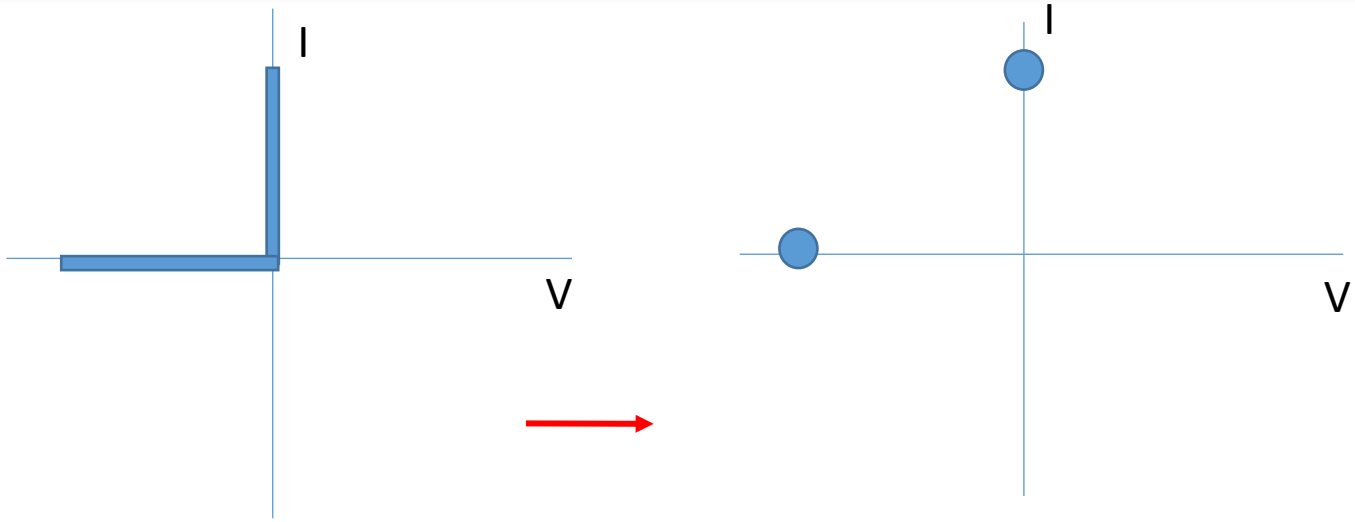


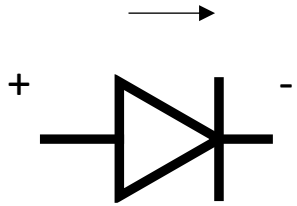


- The same rating is true for the diode.

Quadrants of operation

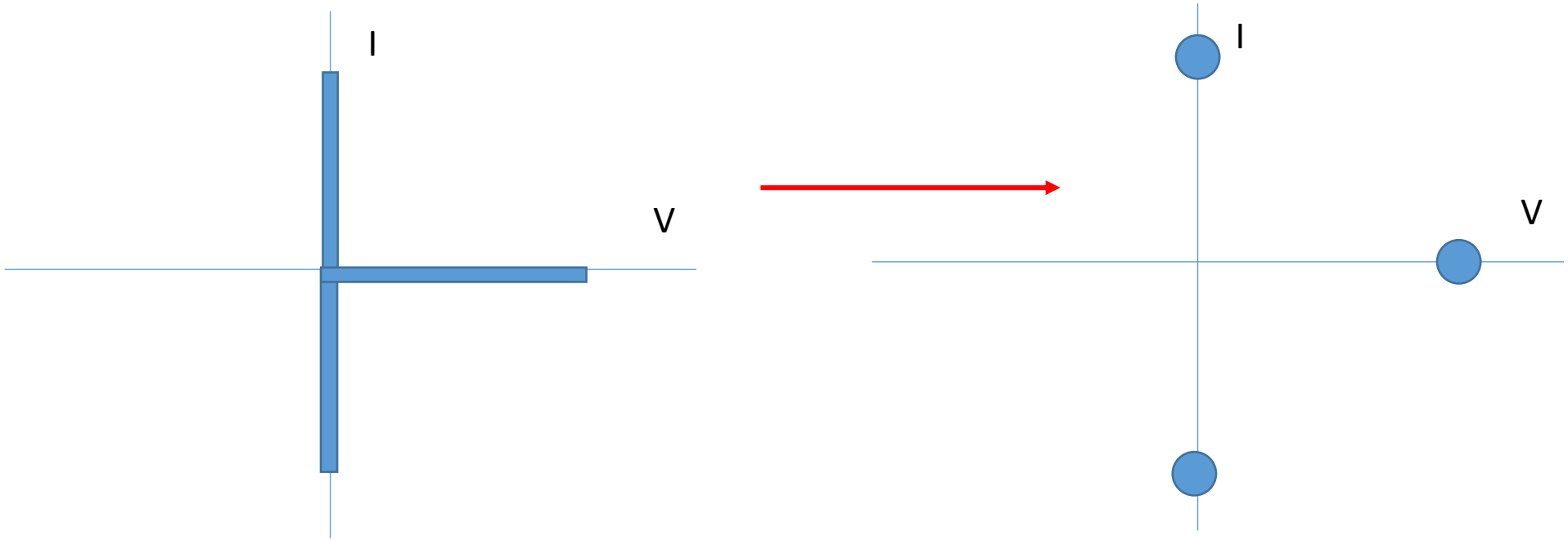


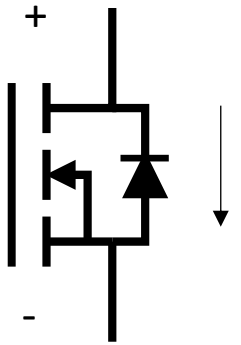




- I stands for current during conduction and V stands for blocking voltage.
- A diode has only 2nd quadrant operation.

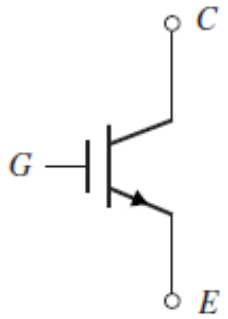
Quadrants of operation

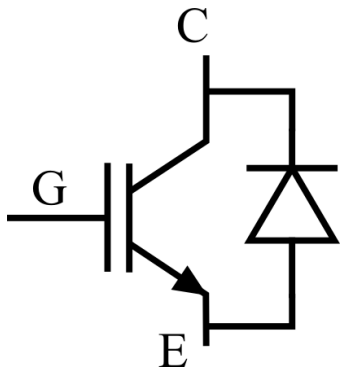




- A MOSFET with anti parallel diode has 2 quadrants of operation.

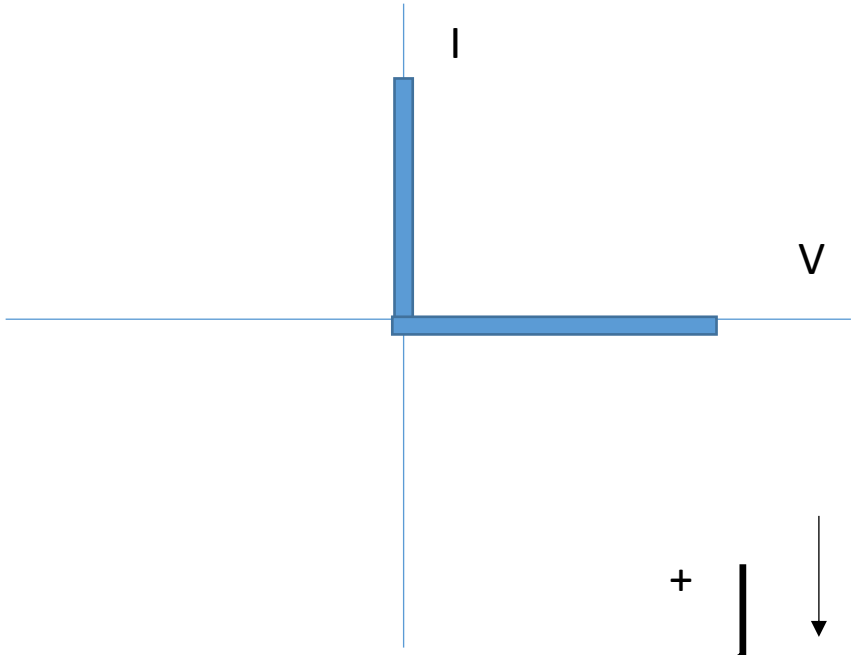
IGBT (Insulated Gate Bipolar Transistor)

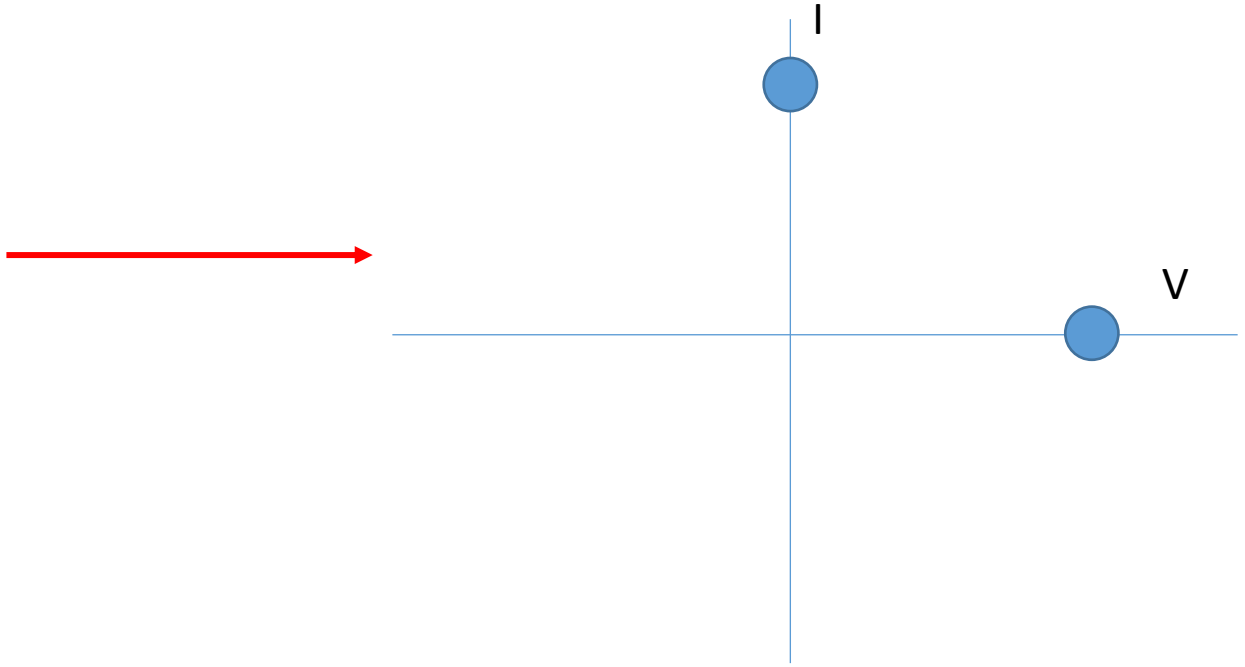


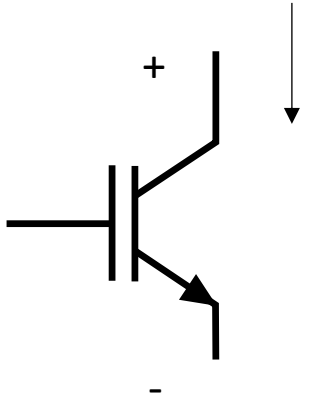


- Combination of MOSFET (low gate drive power requirement, high input impedance) and BJT (high current carrying capacity, low on state loss).
- Available up to 3.3 kV, 1500A.
- Switching frequency up to 20 kHz.
- Mainly used in low and medium power from several hundreds of kW and multi MW level.
- i-v characteristics very similar to MOSFET.

Quadrants of operation

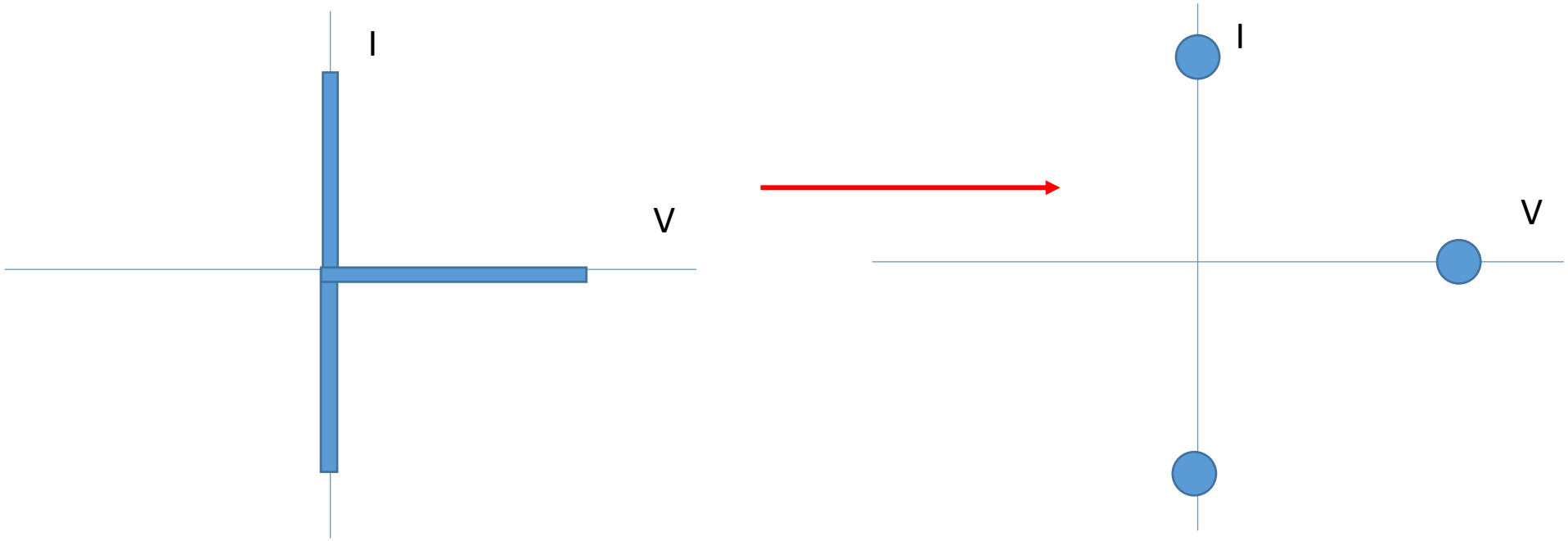


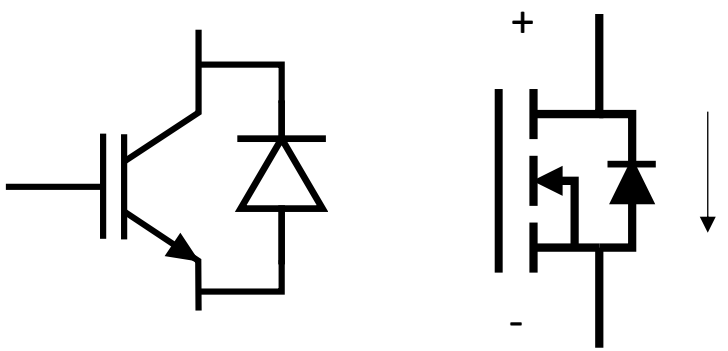




- An IGBT has 1st quadrant operation.

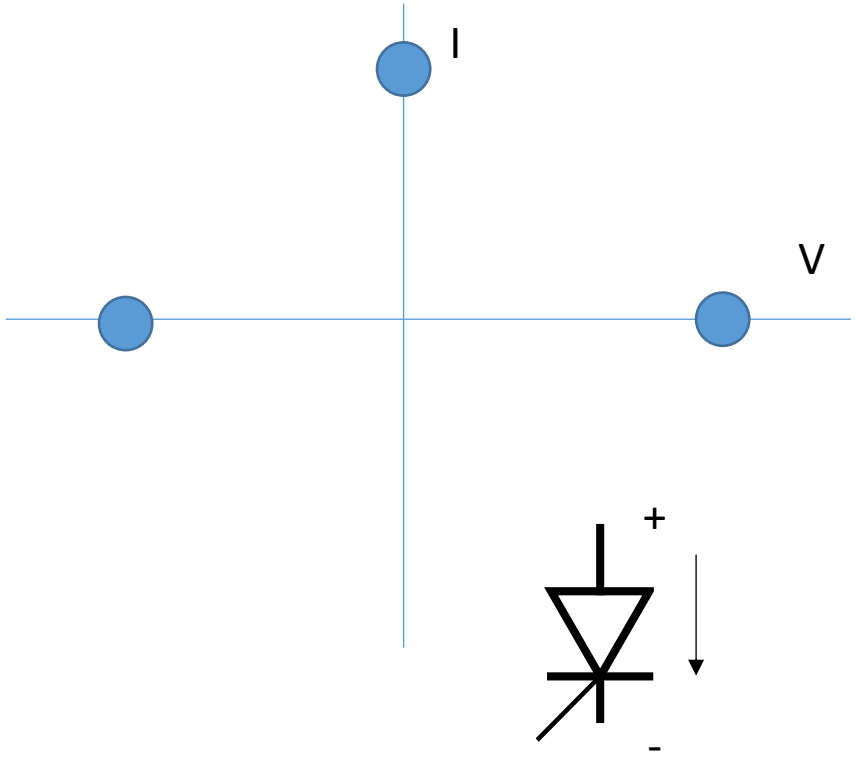
Quadrants of operation





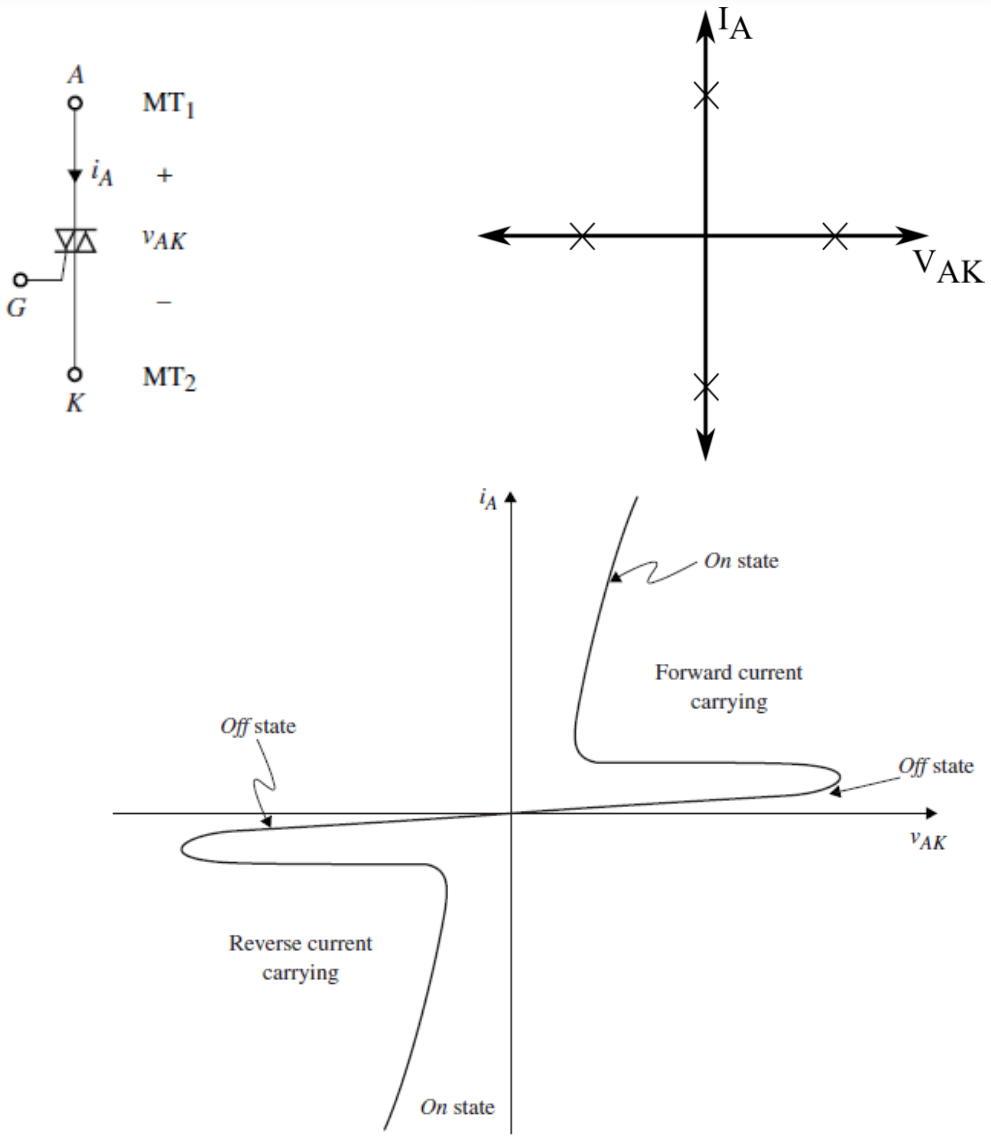
- An IGBT with anti parallel diode has 2 quadrants of operation.

Quadrants of operation



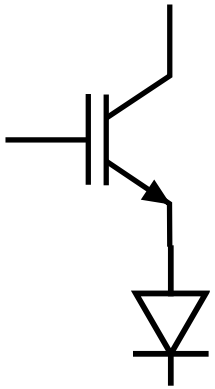
- A thyristor has 2 quadrants of operation.

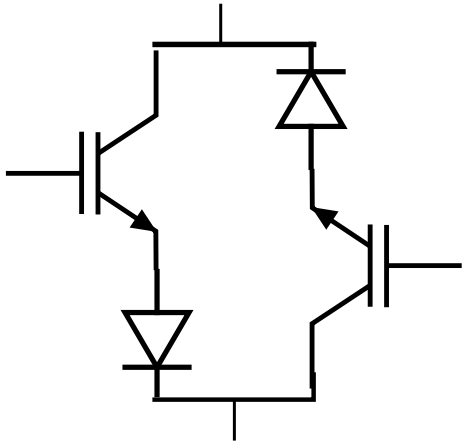
Triac

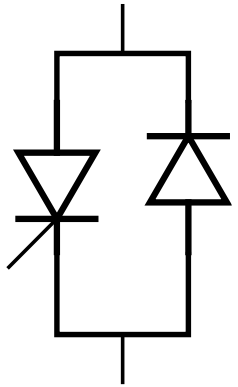


- Triac is a back-to-back Thyristor.
- Used in low power application.
- It is a four-quadrant switch.

Do it yourself







- Find the quadrants of operation of the following configurations.

Important manufacturers

- Several manufacturers like Semikron, Infineon, Mitsubishi, Cree etc. make many power devices of different ratings.
- The datasheet and application notes are a rich source of information. See for example here: https://www.semikron.com/dl/service-support/downloads/download/semikron-application-manual-power-semiconductors-english-en-2015/

Advantages of SiC devices

- Silicon Carbide (SiC) has better electrical properties than Silicon. Compared to Silicon,
  - SiC has higher critical breakdown field (about 10 times), which means the same voltage rating can be maintained while reducing the thickness of the device. With reduction in thickness, the on-state resistance is reduced. Thus conduction losses are smaller.
  - A wider bandgap, leading to lower leakage current at relatively high temperatures. Thus SiC can work at higher operating temperatures.
  - SiC has higher electron saturation velocity (about 2 times), thus the device can turn on and turn off faster.
  - A higher thermal conductivity, which ensures lower temperature rise.

Challenges with SiC devices

- Because SiC can switch faster, the di/dt is faster. This causes higher voltage overshoot and ringing across the device.
- This can cause higher switching losses and EMI issues.
- Packaging is a challenging issue.